RUSHIL BARVE

Computer Science & Computational Mathematics · University of Waterloo



About Me

CS & Computational Mathematics student seeking a Winter/Spring co-op. I ship full-stack projects, lead teams and events, and communicate clearly under pressure.

Contact

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LOCATION
Waterloo, Ontario

Skills

LANGUAGES
Python, JavaScript, TypeScript, GDScript, C, SQL, HTML, CSS
FRAMEWORKS
React, Vite, Tailwind CSS, FastAPI, Godot 4, Zustand
TOOLS
Git, GitHub Actions, SQLite, IndexedDB, Render

Achievements

- Gold Medal of Distinction, International Informatics Olympiad (2023-24); Bronze (2024-25)
- Gold Medal of Distinction, National Science Olympiad
- Six Best Delegate awards across 8+ national and international MUN conferences
- Basketball, CBSE Qatar Clusters: Gold (2025-26), Bronze (2023-24, 2024-25)

Waterloo Teams

- WARG (Waterloo Aerial Robotics): Autonomy and Embedded Flight Software teams
- UW Orbital: student satellite design team
- UW Game Dev · Computer Science Club · UW Data Science Club

EDUCATION

University of Waterloo — Expected 2031
Computer Science (Co-op) · Computational Mathematics

Delhi Public School – Modern Indian School, Doha — Graduated 2026
CBSE Senior Secondary, Computer Science

PROJECTS

CircuitGuard — Python · FastAPI · React · Three.js
Winner, Ampere – AI for Motor Safety · FormulaTech Hacks 2026

- Real-time Formula 1 safety system: analyses live driver telemetry to detect crashes and stopped cars on the racing line and recommends the FIA response (safety car, VSC, red flag) under the 2026 sporting regulations.
- Models per-sector spectator risk with a Bayesian Monte Carlo insurance-pricing model, so premiums can reflect each track section instead of a blanket rate.
- Python/FastAPI backend on the OpenF1 API, FAISS semantic search over FIA regulations, faster-whisper radio transcription, Next.js/React 3D track view.

Whisper Chain — Godot 4 · GDScript
Puzzle game built on function composition

- Puzzle game where each villager garbles a message (reverse, swap, double, drop a word); players route the message so the distortions cancel, discovering inverses and non-commutativity.
- Graph solver (DFS over simple paths) finds every correct route, sets par and powers live hints; used to design and validate all 10 levels.
- ~3,000 lines of GDScript with UI-free game logic, ~1,500 test assertions including randomized property tests, and GitHub Actions CI building web and Windows exports.

MUN Chair Toolkit — TypeScript · React · Vite · Tailwind · Zustand · Dexie
Local-first web app for Model UN chairs · live on GitHub Pages

- Committee-management app for Model UN chairs: committee and agenda setup, bulk delegation entry, and roll call with Present, Present and Voting, and Absent states.
- Automated quorum, simple-majority and two-thirds calculations, so chairs no longer count votes by hand.
- Kept rules in framework-free functions (src/lib/rules.ts) with automated tests; deployed to GitHub Pages.

Door Occupancy Counter (in progress) — Python · SQLite · JavaScript · TensorFlow.js
Prototype: live building occupancy from door sensors

- Counts people entering and leaving each door from two-beam sensor events (direction from beam order), keeping a live headcount with no manual logging.
- Live web dashboard, SQLite event log and automatic daily Excel reports (summary, hourly and event sheets) written with no external libraries.

Source code available **on request** · more projects at rsk0110.github.io/s2rkportfolio

LEADERSHIP & EXPERIENCE

Director of MUN & Electoral and Democratic Club — May 2025 – Jun 2026
DPS-Modern Indian School, Doha

- Led and managed the school's MUN program: organized conferences, trained delegates, promoted diplomacy and public speaking.

Head of Marketing, TEDxYouth@DPSMIS — Sep – Dec 2025

- Led branding, social media and promotion for a TEDx-licensed youth event, working across curation, design, operations and technical teams.

Head of Operations, DPSDQ MUN — Jun – Oct 2025

- Ran logistics and delegate services for the 25th-anniversary edition, coordinating with the Secretariat.

Team Leader, Qatar Children's Science Congress — Dec 2024 – Mar 2025

- Led an original student research project from experimental design to data analysis and expert-panel presentation.

HACKATHONS

FormulaTech Hacks — Winner (Ampere)	2026	RevenueCat Shipaton	2026
OpenCV AI Competition (AWS)	2026	Nebius x NVIDIA Global AI Hackathon	2026
IBM BOB 2.0 Hackathon	2026	DPS MIS Doha Hackathon	2025
Birla School Hackathon — Qatar	2024	CMU Qatar Hackathon	2023

8 hackathons · 5 won · 1 organised